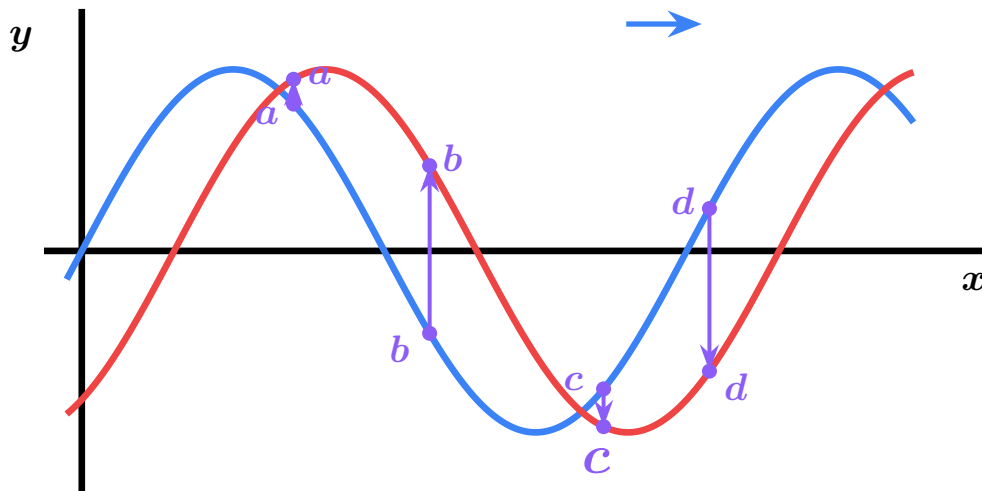


The Wave Equation and Travelling Solutions

1 Up-And-Down, There Goes the Wave

(a)

The blue line is the curve at time t , and the red line is the wave at time $t + \Delta t$.



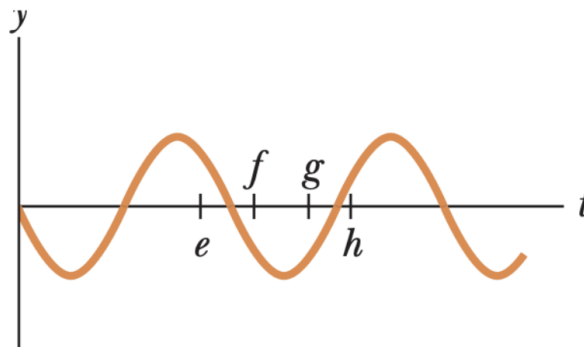
So at the instant when time is t :

- a, b points move **upward**
- c, d points move **downward**

(b)

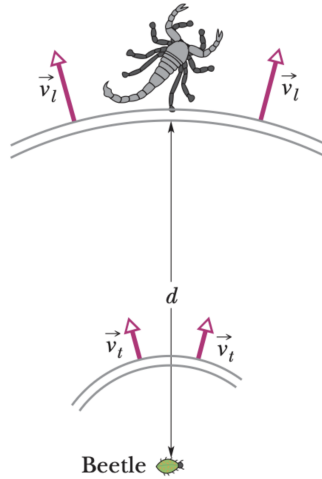
If $\frac{dy}{dt} > 0$ (vertical velocity of a particle on the wave), then the string element at that instant moves upwards.

If $\frac{dy}{dt} < 0$, the string element at that instant moves downwards.



- e, f move **downward**
- g, h move **upward**

2 The Beetle and The Scorpion



(a)

Time for longitudinal wave to reach scorpion $= t_l = \frac{d}{v_l}$

Time for transverse wave to reach scorpion $= t_t = \frac{d}{v_t}$

$d \rightarrow$ distance between scorpion and beetle.

And

$$\Delta t = t_t - t_l = d \left(\frac{1}{v_t} - \frac{1}{v_l} \right)$$

$$\Rightarrow \Delta t = d \left(\frac{1}{v_t} - \frac{1}{v_l} \right)$$

$$\Rightarrow \boxed{d = \Delta t \frac{v_t v_l}{v_l - v_t}}$$

(b)

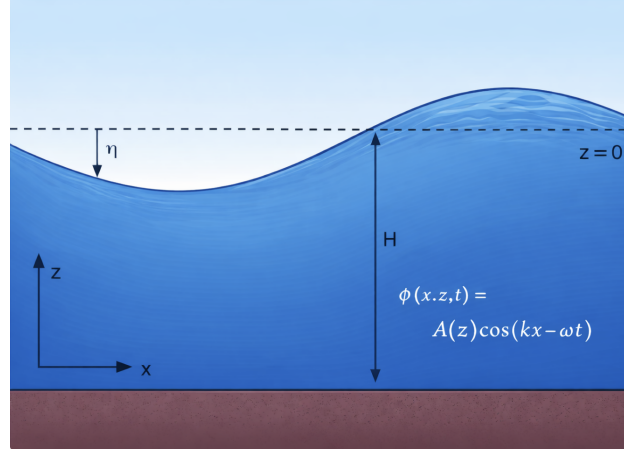
$$v_t = 50 \text{ m/s}, \quad \Delta t = 4 \text{ ms}, \quad v_l = 150 \text{ m/s}$$

$$d = 4 \times 10^{-3} \cdot \frac{50 \times 150}{100}$$

$$\Rightarrow \boxed{d = 0.3 \text{ m} = 30 \text{ cm}}$$

The scorpion would have to jump 30 cm.

3 Ocean Waves in Deep Water



(a)

$$v_x = \frac{\partial \phi}{\partial x} = \frac{\partial}{\partial x} [A(z) \cos(kx - \omega t)]$$

$$\Rightarrow v_x = -k A(z) \sin(kx - \omega t)$$

$$v_z = \frac{\partial \phi}{\partial z} = \frac{\partial}{\partial z} [A(z) \cos(kx - \omega t)] = \frac{\partial A(z)}{\partial z} \cos(kx - \omega t)$$

(b)

$$\frac{\partial^2 \phi}{\partial x^2} = \frac{\partial v_x}{\partial x} = \frac{\partial}{\partial x} [-k A(z) \sin(kx - \omega t)] = -k^2 A(z) \cos(kx - \omega t)$$

and

$$\frac{\partial^2 \phi}{\partial z^2} = \frac{\partial v_z}{\partial z} = \frac{\partial}{\partial z} \left[\frac{\partial A(z)}{\partial z} \cos(kx - \omega t) \right] = \frac{\partial^2 A(z)}{\partial z^2} \cos(kx - \omega t)$$

Thus $\frac{\partial^2 \phi}{\partial z^2} + \frac{\partial^2 \phi}{\partial x^2} = 0$ gives:

$$\left(\frac{\partial^2 A(z)}{\partial z^2} - k^2 A(z) \right) \cos(kx - \omega t) = 0$$

$$\Rightarrow \left(\frac{\partial^2 A(z)}{\partial z^2} - k^2 A(z) \right) = 0$$

(c)

Assume $A(z) = C e^{mz}$.

Substituting into the differential equation, we find:

$$C m^2 e^{mz} - k^2 C e^{mz} = 0$$

$$\Rightarrow m^2 - k^2 = 0$$

$$\Rightarrow \boxed{m = \pm k}$$

Taking into account the linear nature of ODEs and their solutions:

$$\boxed{A(z) = C_1 e^{-kz} + C_2 e^{kz}} \quad (C_1, C_2 \text{ are constants})$$

(d)

At infinite depth $\phi \rightarrow 0$ as motion is negligible.

That means as $z \rightarrow -\infty$, $\phi \rightarrow 0 \Rightarrow A(z) \rightarrow 0$.

For this to be possible, $C_1 = 0$.

Thus e^{-kz} is *not* a solution.

And

$$\boxed{A(z) = A_0 e^{kz}} \quad (A_0 = A(0))$$

(e)

$$\eta(x, t) = B \sin(kx - \omega t)$$

$$\left. \frac{d\phi}{dz} \right|_{z=0} = \frac{d\eta}{dt}$$

$$\Rightarrow \left. \frac{dA(z)}{dz} \cos(kx - \omega t) \right|_{z=0} = -\omega B \cos(kx - \omega t)$$

$$\Rightarrow A_0 \left. \frac{d e^{kz}}{dz} \right|_{z=0} \cos(kx - \omega t) = -\omega B \cos(kx - \omega t)$$

$$\Rightarrow A_0 k = -\omega B$$

$$\Rightarrow \boxed{B = -\frac{k}{\omega} A_0}$$

(f)

$$\frac{\partial \phi}{\partial t} + g\eta = 0 \quad \text{at } z = 0$$

$$\frac{d\phi}{dt} + g\eta = \omega A(z) \sin(kx - \omega t) + g B \sin(kx - \omega t) = (\omega A_0 e^{kz} + g B) \sin(kx - \omega t)$$

At $z = 0$:

$$\omega A_0 + g B = 0$$

$$\Rightarrow \omega = -\frac{g B}{A_0}$$

$$\Rightarrow \omega = \frac{gk}{\omega} \quad \left(\text{using } B = -\frac{k}{\omega} A_0 \right)$$

$$\Rightarrow \omega^2 = gk$$

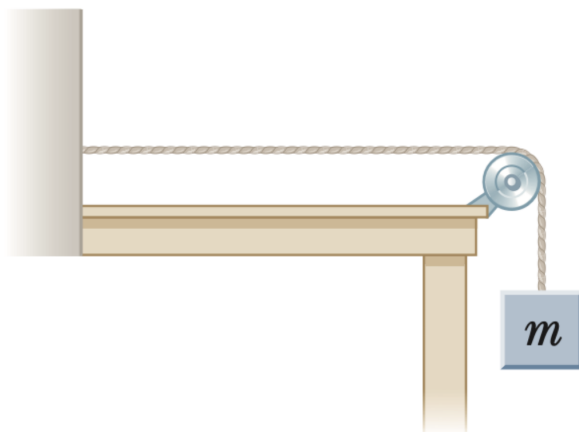
$$\Rightarrow \boxed{\omega = \sqrt{gk}}$$

(g)

We know $v = \frac{\omega}{k}$

$$\Rightarrow v = \frac{\sqrt{gk}}{k} = \sqrt{\frac{g}{k}} = \sqrt{\frac{g\lambda}{2\pi}} \quad \left(\text{since } k = \frac{2\pi}{\lambda} \right)$$

4 More Waves on Strings, on Earth and on the Moon



(a)

Let's ignore the mass of the string that is hanging off the pulley.

Then $T = mg$ and $\mu = \frac{m_{\text{st}}}{l}$.

We know

$$v = \sqrt{\frac{T}{\mu}} = \sqrt{\frac{mgl}{m_{\text{st}}}} \quad \leftarrow \text{Speed of a wave on the string}$$

(b)

We know that $f\lambda = v \quad \dots (1)$

and $v = \sqrt{\frac{mgl}{m_{\text{st}}}}$

Then using (1), we get

$$\lambda = \frac{v}{f} = \frac{1}{f} \sqrt{\frac{mgl}{m_{\text{st}}}}$$

$$\Rightarrow \boxed{\lambda = \frac{1}{f} \sqrt{\frac{mgl}{m_{\text{st}}}}}$$

(c)

Since the mass of the string hanging off the pulley is tiny compared to 2 kg, we can ignore its contribution to the tension.

Thus $T = 2g$ ($g \rightarrow$ acceleration due to gravity)

and $\mu = \frac{m_{\text{st}}}{l} = \frac{4 \times 10^{-3}}{1.6} = 2.5 \times 10^{-3} \text{ kg/m}$

Then

$$v = \sqrt{\frac{2g}{2.5 \times 10^{-3}}} = \sqrt{\frac{2}{2.5} \times 10^2} \cdot \sqrt{\frac{2}{2.8}} \quad (\text{take } g = 10 \text{ m/s}^2) = \frac{2}{\sqrt{5}} \times 10^2 \text{ m/s}$$

Then the time taken for the wave to cover the horizontal portion of the string is:

$$t = \frac{l}{v} = \frac{1.3}{\frac{2}{\sqrt{5}} \times 10^2} = 14.53 \text{ ms}$$

$$\boxed{t = 14.53 \text{ ms}}$$

(d)

If this experiment was carried out on the moon, then

$$g' = g/6 \quad (g' \rightarrow \text{Acceleration due to gravity on the moon})$$

Then

$$v' = \sqrt{\frac{mg'l}{m_{\text{st}}}} = \frac{1}{\sqrt{6}} \sqrt{\frac{mgl}{m_{\text{st}}}} = \frac{v}{\sqrt{6}}$$

$v' \rightarrow$ velocity of wave on the moon.

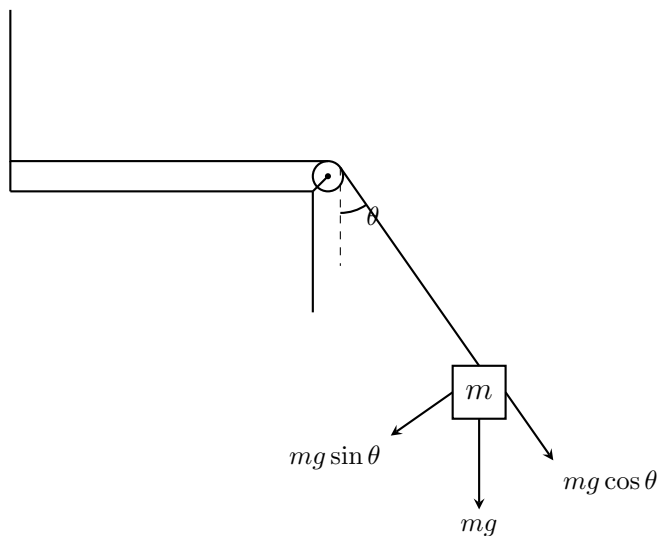
and

$$\begin{aligned} t' &= \frac{l}{v'} = \sqrt{6} \frac{l}{v} = \sqrt{6} t \\ \Rightarrow t' &= \sqrt{6} t \end{aligned}$$

$t' \rightarrow$ time taken for the wave to travel the horizontal distance on the moon.

(e)

If the block was swaying like a pendulum, the tension would vary with θ .



$$T = mg \cos \theta$$

which would mean

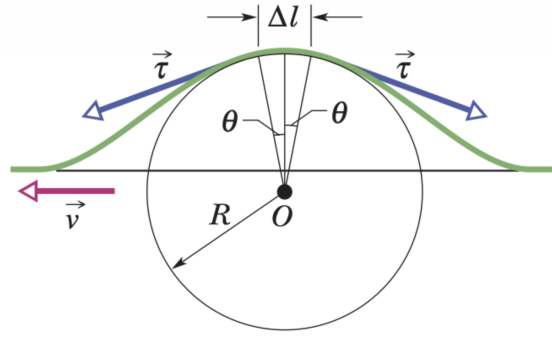
$$v = \sqrt{\frac{mgl \cos \theta}{m_{\text{st}}}}$$

Velocity would also vary as a function of θ .

5 Waves on a Rope [Experiment]

Measurement has to be carried out and data analyzed.

6 The Wave Speed, in a Different Way



(a)

The horizontal components cancel each other out.

Thus $F_H = 0$.

The vertical component is

$$F_V = 2\tau \sin \theta$$

(b)

Since θ is small, $\sin \theta \approx \tan \theta \approx \frac{\Delta l}{2R}$

$$\Rightarrow F_V \approx 2\tau \cdot \frac{\Delta l}{2R} = \frac{\tau \Delta l}{R}$$

$$\Rightarrow F_V \approx \frac{\tau \Delta l}{R}$$

(c)

Let the mass of the small element be Δm .

Centrifugal force on the mass element (F_c) = $\frac{\Delta m v^2}{R}$

and $F_c = F_V$

$$\Rightarrow \frac{\Delta m v^2}{R} = \frac{\tau \Delta l}{R}$$

$$\Rightarrow v^2 = \tau \frac{\Delta l}{\Delta m} = \frac{\tau}{\Delta m / \Delta l} = \frac{\tau}{\mu}$$

$$\Rightarrow \boxed{v = \sqrt{\frac{\tau}{\mu}}}$$